Report: Example D2

Date **2026-11-03** Engine **FAKE fake-2** App **0.1.0**

> ⚠ **FAKE engine:** These results come from the Fake Engine. They are placeholders, not engineering results.

1 Verdict

✕ **INFEASIBLE**: A valid operating point exists, but the design violates 2 constraint(s).

2 Limiting constraints

- Bus voltage, lower limit at B1: 0.9 pu against the limit 1 pu.
- Inverter loading at INV1: 1.1 ratio against the limit 1 ratio.

Limiting constraints

Constraint	Element	Value	Limit	Unit	Margin
Bus voltage, lower limit	B1	0.9	1	pu	✕ **Violated by 0.1 pu**
Inverter loading	INV1	1.1	1	ratio	✕ **Violated by 0.1 ratio**

3 Margins and checks not evaluated

All evaluated margins

Constraint	Element	Value	Limit	Unit	Margin
Bus voltage, lower limit	B1	0.9	1	pu	✕ **Violated by 0.1 pu**
Inverter loading	INV1	1.1	1	ratio	✕ **Violated by 0.1 ratio**

Not evaluated

- Bus voltage, lower limit, B1: no limit given; not counted as satisfied.
- Bus voltage, upper limit, B1: no limit given; not counted as satisfied.
- Bus voltage, lower limit, B2: no limit given; not counted as satisfied.
- Bus voltage, upper limit, B2: no limit given; not counted as satisfied.
- Bus voltage, lower limit, B3: no limit given; not counted as satisfied.
- Bus voltage, upper limit, B3: no limit given; not counted as satisfied.
- Frequency, lower limit, SYSTEM: no limit given; not counted as satisfied.
- Frequency, upper limit, SYSTEM: no limit given; not counted as satisfied.
- Inverter loading, INV1: no limit given; not counted as satisfied.
- Inverter loading, INV2: no limit given; not counted as satisfied.
- Line current, LN1: no limit given; not counted as satisfied.
- Line current, LN2: no limit given; not counted as satisfied.

Loadability

λ_c (constraint loadability, ratio)	not available
λ_s (ratio)	not available

solution existence, not a stability limit

Resolution (ratio)	0.005
Grid step (ratio)	0.05
Stop reason	FAKE_ENGINE

ADVICE **Sizing check (advisory, from the network)**

Inverter	S (kVA)	Required rating (kVA)	Headroom (kVA)	Rating margin short
INV1	0	0	70	no
INV2	0	0	45	no

Line	Current (A)	Required current (A)
LN1	0	0
LN2	0	0

No storage check. Fake Engine placeholder; not a calculation result.

4 Design summary

Quantity	Value	Unit
Nominal voltage, line-to-line	0.44	kV
Nominal frequency	50	Hz
Buses	3	count
Lines	2	count
Loads	1	count
Inverters	2	count

5 Results and charts



Bus voltage magnitude (p.u.) from the latest power flow (M1).

System

Quantity	Value	Unit
System frequency	50	Hz
Total load, active	0	kW
Total load, reactive	0	kvar
Total losses	0	kW
Iterations	0	count
Validity indicator ε_X	0	ratio

Bus voltages

Bus	**Voltage** (kV)	**Voltage** (p.u.)	**Angle** (deg)
B1	0.44	1	0
B2	0.44	1	0
B3	0.44	1	0

Inverter outputs

Inverter	**P** (kW)	**Q** (kvar)	**S** (kVA)	**Loading** (ratio)
INV1	0	0	0	0
INV2	0	0	0	0

ADVICE **Power sharing against rating-proportional sharing (advisory, not a constraint)**

Inverter	**ΔP** (kW)	**ΔQ** (kvar)	**P error** (pu of rating)	**Q error** (pu of rating)	**Q–V droop not rating-proportional**
INV1	0	0	0	0	no
INV2	0	0	0	0	no

Line flows

Line	**P from** (kW)	**Q from** (kvar)	**P to** (kW)	**Q to** (kvar)	**Current** (A)	**Losses** (kW)
LN1	0	0	0	0	0	0
LN2	0	0	0	0	0	0

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6 Assumptions

Sizing check: Fake Engine placeholder; not a calculation result.

Validity indicators: threshold not yet calibrated

Indicator	Checks assumption	Value (ratio)	From
r_τ line-to-filter time ratio (Eq. 4.23)	A1	not evaluated: run small-signal (M4a) on the current design	none
κ dominant-mode speed (Eq. 4.23)	A1	not evaluated: run small-signal (M4a) on the current design	none
r_in inner-loop ratio (Eq. 4.23)	A2	not evaluated: run small-signal (M4a) on the current design	none
ε_X validity indicator (Eq. 4.1)	A4	0	M1 run e528624497404fd48e6c370c464ed73e (Fake Engine)

Model assumptions (manuscript Tabel 1.1) and how each is handled

Assumption	Statement	Handling	Indicator values	RunRecord result hash
A1	Jaringan kuasi-statis (saluran dimodelkan aljabar)	Checked by indicator (r_τ, κ)	r_τ: not evaluated: run small-signal (M4a) on the current design; κ: not evaluated: run small-signal (M4a) on the current design	
A2	Inverter sebagai sumber tegangan ideal inner loop dalam, filter LC, PLL, dan pembatas arus	By construction; checked by indicator (r_in)	r_in: not evaluated: run small-signal (M4a) on the current design	
A3	Konstanta waktu filter daya ditetapkan pengguna	Set by the user; explored by sweep (M3b)		
A4	Reaktansi saluran tidak diperbarui terhadap frekuensi; tanpa shunt	Checked by indicator (ε_X)	ε_X: 0	`4ff7a438f42d1cb61c2ad3ec6ea370d93a1e05c14494b078c77cfa27b48067bd`
A5	Kendali droop primer saja	By construction		
A6	Tiga fasa seimbang	By construction		

7 Traceability

Analysis	RunRecord id	Snapshot hash	Result hash
M1	e528624497404fd48e6c370c464ed73e	`149741dac4f0e99f80aa65b8353605b957f1e08ba2402395ad11f319e12b1288`	`4ff7a438f42d1cb61c2ad3ec6ea370d93a1e05c14494b078c77cfa27b48067bd`
M2	50387504b1c1466db7f97d2ddb171607	`149741dac4f0e99f80aa65b8353605b957f1e08ba2402395ad11f319e12b1288`	`fddd9d67397b677b172b87de8e8a1acc8cc177b76a8bc88d35a55dd6cfc12c33`